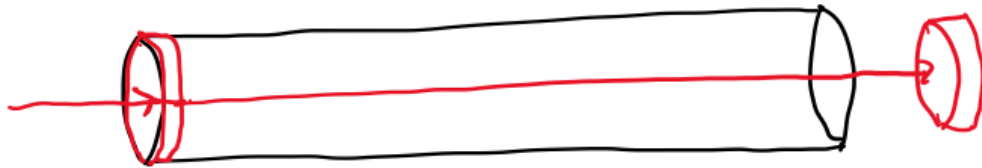


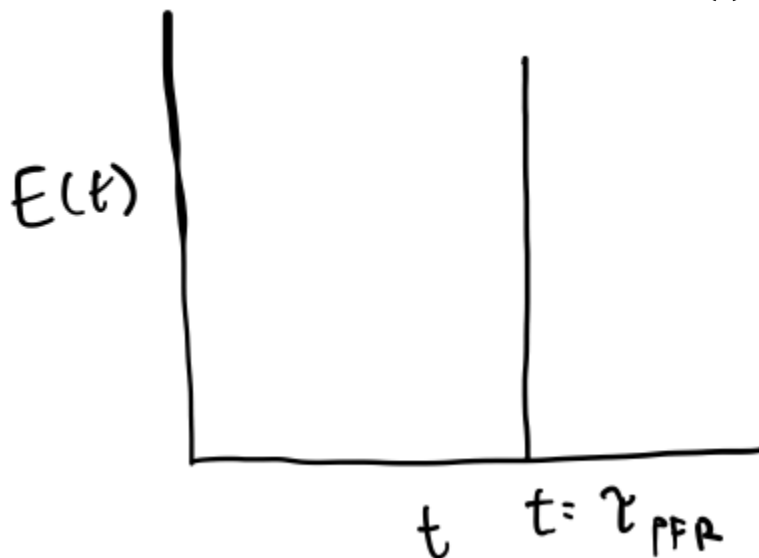
27 Residence time distribution in ideal PFRs and CSTRs

- We discussed in a prior lecture that there is only one value for residence time in an ideal PFR/PBR since

there is perfect mixing in the radial dimension & no mixing in the axial dimension. So all the fluid takes the same amount of time to travel from the inlet to the outlet.

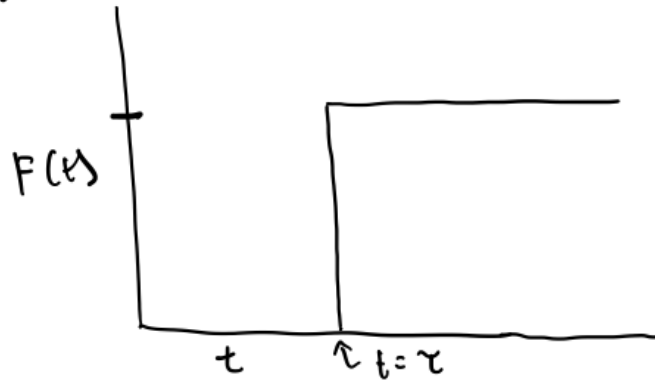


- The residence time distribution $E(t)$ hence looks like



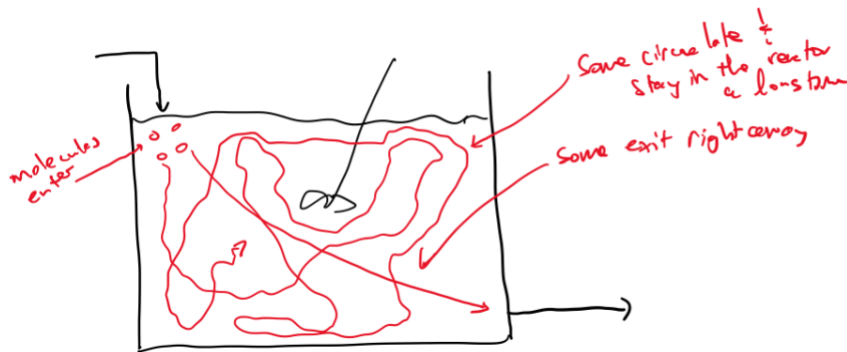
- The cumulative distribution $F(t)$ looks like

$F(t)$ tells the fraction of molecules that have been in the reactor for time less than t



- We discussed previously that CSTRs, even ideal CSTRs, exhibit a distribution of residence times since

Some molecules will exit the reactor quickly, while others will remain in the reactor for a long time



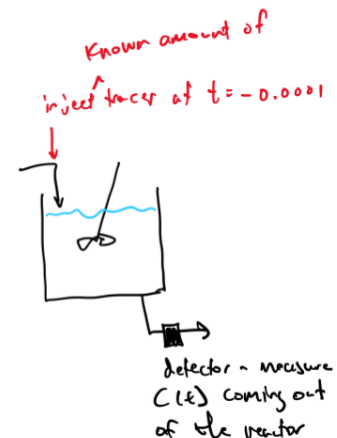
- Consider a pulse experiment in a CSTR. We are going to write the mole balance on the tracer for the instant after the pulse was injected:

Accumulation = In - Out + Generation - Consumption

$$\frac{dN_A}{dt} = 0 - v C_A + 0 - 0$$

$$\frac{d(V C_A)}{dt} = -v C_A$$

- Assume liquid conditions
($v = \text{constant}$), also
constant V



$$V \frac{dC_A}{dt} = -r C_A \rightarrow \tau \frac{dC_A}{dt} = -C_A \rightarrow \frac{dC_A}{C_A} = -\frac{1}{\tau} dt$$

of the reactor

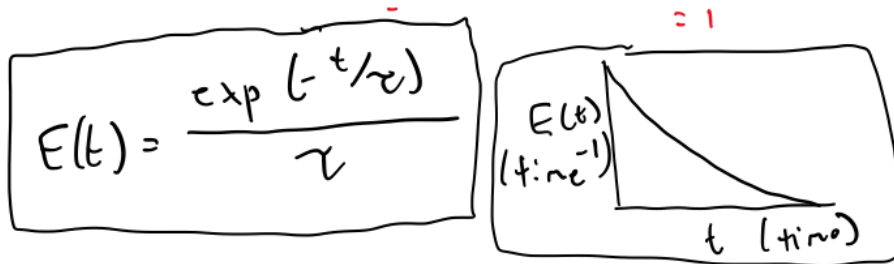
$$\int_{C_{A0}}^{C_A} \frac{dC_A}{C_A} = -\frac{1}{\tau} \int_0^t dt$$

$$C_A(t) = C_{A0} \exp(-t/\tau)$$

$$E(t) = \frac{C(t)}{\int_0^\infty C(t) dt} = \frac{C_{A0} \exp(-t/\tau)}{C_{A0} \int_0^\infty \exp(-t/\tau) dt}$$

$$\int_0^\infty \exp(-t/\tau) dt \approx \lim_{t \rightarrow \infty} \int_0^t \exp(-t/\tau) dt$$

$$= -\tau \left[\underbrace{\lim_{t \rightarrow \infty} \exp(-t/\tau)}_{=0} - \underbrace{\exp(-t/\tau)}_{=1} \right]_{t=0} = \tau$$



$$\int_0^\infty E(t) dt = 1$$

- The dimensionless RTD is given by $E(\theta)$, where $\theta = t/\tau$

- $E(\theta)$ is the same for any ideal CSTR. $E(\theta) = \exp(-\theta) = \tau E(t)$

- $F(\theta)$ for an ideal CSTR:

$$F(\theta) = \int_0^\theta E(\theta) d\theta = \int_0^\theta \exp(-\theta) d\theta = 1 - \exp(-\theta)$$

- t_m for an ideal CSTR $\hat{=} \tau$

$$t_m = \int_0^{\infty} t E(t) dt = \int_0^{\infty} \frac{t}{\tau} \exp(-t/\tau) dt = \frac{1}{\tau} \int_0^{\infty} t \exp(-t/\tau) dt$$

integration by parts

$$\int u dv = uv - \int v du \quad \begin{array}{ll} u = t & dv = \exp(-t/\tau) dt \\ du = dt & v = -\tau \exp(-t/\tau) \end{array}$$

$$\begin{aligned} \int t \exp(-t/\tau) dt &= -t\tau \exp(-t/\tau) + \tau \int \exp(-t/\tau) dt \\ &= -t\tau \exp(-t/\tau) - \tau^2 \exp(-t/\tau) \end{aligned}$$

$$\begin{aligned} \frac{1}{\tau} \int_0^{\infty} t \exp(-t/\tau) dt &= \lim_{t \rightarrow \infty} t \exp(-t/\tau) - \tau \lim_{t \rightarrow \infty} \exp(-t/\tau) \\ &= \left[\underset{\uparrow = 0}{0} \cdot \exp(0) - \tau \cdot \underset{\uparrow = 1}{\exp(0)} \right] \end{aligned}$$

$$\frac{1}{\tau} \int_0^{\infty} t \exp(-t/\tau) dt = \tau$$

So, for an ideal CSTR, $t_m = \tau_{CSTR}$.